MALLA REDDY
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Multi Agentic Workflow for student Attendance Recognition and Real-time Monitoring LMS

ABSTRACT

Multi Agentic Workflow for student Attendance Recognition and Real-time Monitoring LMS is an AI-powered classroom monitoring and engagement analysis system designed to enhance the effectiveness of online learning environments through real-time computer vision and multi-agent intelligence. The system goes beyond traditional attendance tracking by continuously analyzing student presence, attentiveness, and identity during live virtual sessions. Using facial detection, facial landmark analysis, and head-pose estimation techniques, Multi Agentic Workflow for student Attendance Recognition and Real-time Monitoring LMS verifies whether enrolled students remain present and actively engaged throughout a class. Independent AI agents collaboratively monitor attendance, detect signs of disengagement, and identify potential proxy attendance or identity changes, thereby supporting academic integrity. Built using technologies such as OpenCV, MediaPipe, Python, and modern web frameworks, the system operates on top of existing video-conferencing platforms without requiring complex infrastructure modifications. Multi Agentic Workflow for student Attendance Recognition and Real-time Monitoring LMS aims to provide educators with meaningful insights into student participation while maintaining a human-in-the-loop approach that minimizes bias, false positives, and privacy concerns. The resulting solution demonstrates the practical application of computer vision and multi-agent AI in creating more interactive, secure.

Student's Name	Roll No	Signature
Reviewer Comments		

Guide Name & Sign **Reviewer Sign** **AD-Coordinator Sign**